

DESIGN AND FABRICATION OF QUADRUPED ROBOT FOR SEARCH AND RESCUE OPERATION

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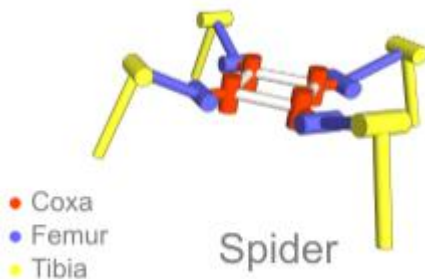
Project Objectives

➤ Need for Simple and Affordable Quadruped Robotic Platforms

- ❖ There is a growing demand for low-cost, easy-to-build quadruped robots that can support hands-on learning, prototyping, and experimentation in embedded systems and legged locomotion.
- ❖ Quadruped robots provide better stability and adaptability on uneven surfaces compared to wheeled platforms, making them suitable for educational and exploratory applications.

➤ Practical Challenges in Implementing Low-Power Quadruped Systems

- ❖ Most quadruped robots rely on complex control algorithms and high computational resources, which limits their use on low-power microcontrollers in cost-sensitive environments.
- ❖ Achieving smooth locomotion, reliable wireless control, and basic obstacle avoidance using simple sensors and embedded hardware remains a significant design challenge.



Existing Techniques

- ❖ **Rahman et al. (2023)** present a low cost 12 DoF quadruped robot design that balances affordability with functional locomotion, enabling easier research and educational adoption
- ❖ **Nguyen et al. (2025)** propose a gait planning method using IMU data and an Augmented Random Search (ARS) algorithm to achieve stable locomotion on varied slopes, demonstrating low cost sensor utility for terrain adaptation.
- ❖ **Hoang et al. (2024)** explore obstacle avoidance control in quadruped robots using the potential functions method, integrating camera based recognition and algorithmic path planning for safe navigation.
- ❖ **Al-Ayen et al. (2024)** reviewed on synthesizes quadruped robot designs, gait generation techniques, and dynamic stability insights, framing state of the art developments and research gaps.

Proposed Method

- ❖ Utilizes affordable 3D-printed parts, low-power microcontrollers, and basic sensors, enabling hands-on prototyping and community-level robotics education, unlike most high-cost alternatives.
- ❖ Introduces a simple, iterative angle tuning method for servo-controlled leg articulation, achieving stable multi-terrain locomotion without complex inverse kinematics or high computational load.

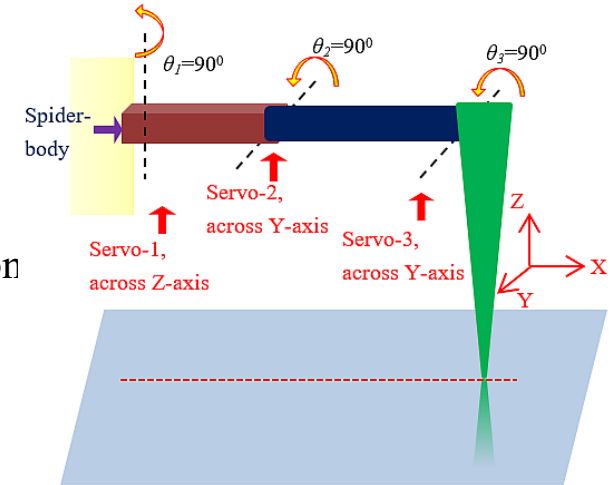
Project Novelty

- ❖ **Adaptive & Stable Locomotion System** – The robot uses bio-inspired gait patterns with real-time servo coordination to achieve smooth, stable movement on uneven terrains, including walking, crawling, and turning actions.
- ❖ **Energy-Efficient and Lightweight Design** – Designed with optimized gait planning and a compact, lightweight structure, the robot ensures reduced power consumption while maintaining flexibility and speed.
- ❖ **Modular and Low-Cost Architecture** – Built using affordable components like Arduino and SG90 servos, with a modular leg design that allows easy control, maintenance, and part replacement.
- ❖ **Smart Control and Future Expandability** – Supports wireless control and provides scope for integrating sensors, obstacle detection, and AI-based features for advanced autonomous applications.

Methodology — System Architecture

➤ Hardware Design

- ❖ 3D-printed quadruped frame for lightweight, low-cost construction.
- ❖ Arduino Nano microcontroller for low-power computation.
- ❖ SG-90 servo motors for multi-degree-of-freedom leg articulation
- ❖ Ultrasonic sensor for obstacle detection and terrain adaptation.
- ❖ Bluetooth module for wireless control.



➤ Sensor & Actuator Integration

- ❖ Servos calibrated using iterative, trial-and-error tuning.
- ❖ Sensors provide real-time feedback for obstacle avoidance and gait correction.
- ❖ Wireless interface allows remote locomotion control (forward, backward, turn).



Methodology — Gait Planning & Control

➤ Leg Motion Strategy

- ❖ Each leg controlled independently for lift, swing, and stance phases.
- ❖ Gait sequence designed for stability on uneven terrain.

➤ Heuristic Calibration

- ❖ Servo angles fine-tuned manually using iterative testing.
- ❖ Motion will be further adjusted based on obstacle detection.

➤ Experimental Validation

- ❖ Tested on flat and slightly uneven surfaces to evaluate:
- ❖ Locomotion stability.
- ❖ Directional control.
- ❖ Obstacle avoidance efficiency. (*future scope*)

➤ Novelty Highlights

- ❖ Avoids high computational algorithms like inverse kinematics.
- ❖ Integrates symmetry and articulated motion for stable multi-joint coordination.
- ❖ Accessible for community-level and educational implementation

Results — Locomotion Performance & Control

➤ Multi-Terrain Locomotion

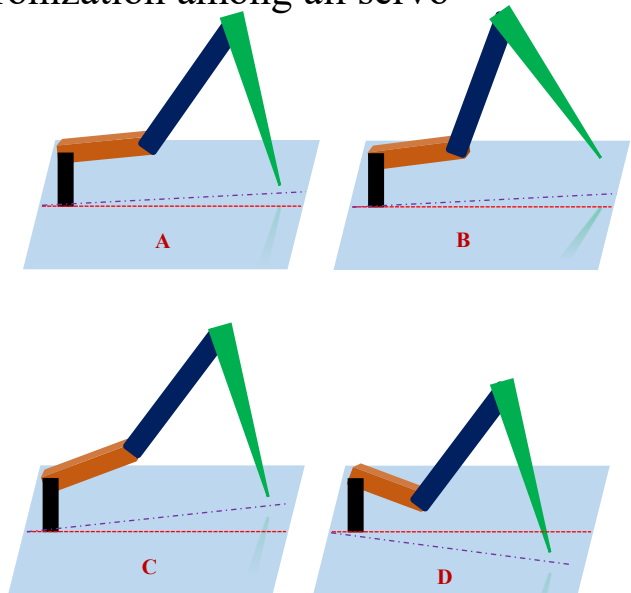
- ❖ The quadruped robot demonstrated stable walking on both flat and uneven terrains with minimal body oscillation.
- ❖ Symmetric gait sequencing ensured balance without complex kinematic or optimization algorithms, reducing computational overhead.
- ❖ The angular orientations of the FR and BL legs are identical.
- ❖ The angular positions of FL and BR legs are concomitant.

➤ Directional Motion Control

- ❖ Reliable forward, backward, left, and right movements were achieved through wireless control commands. (*future scope*)
- ❖ Motion transitions were smooth, indicating effective synchronization among all servo-driven legs.

➤ Obstacle Detection Response (*future scope*)

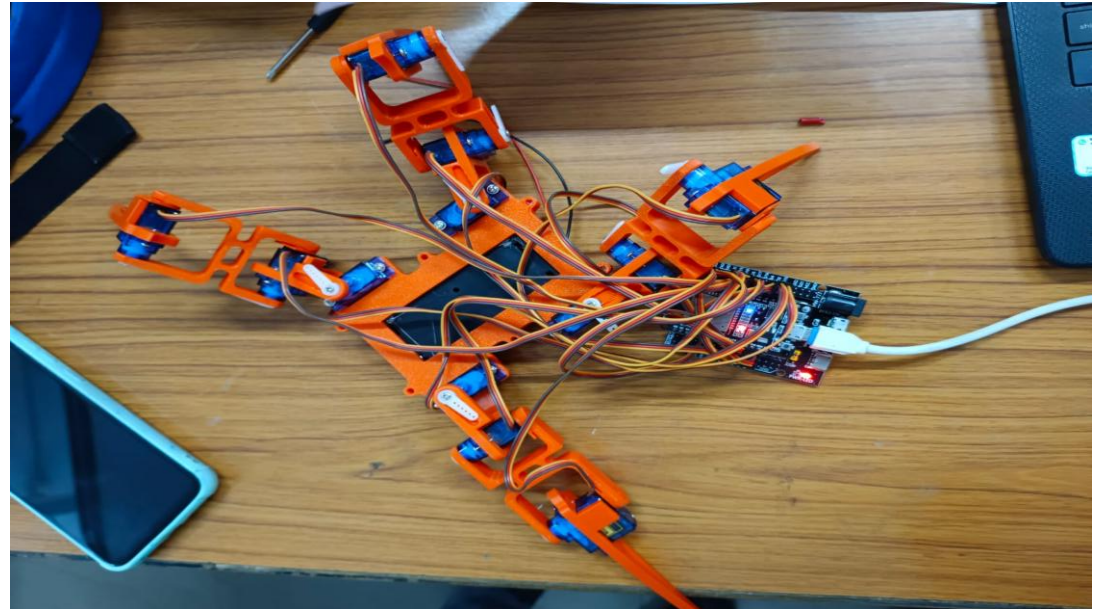
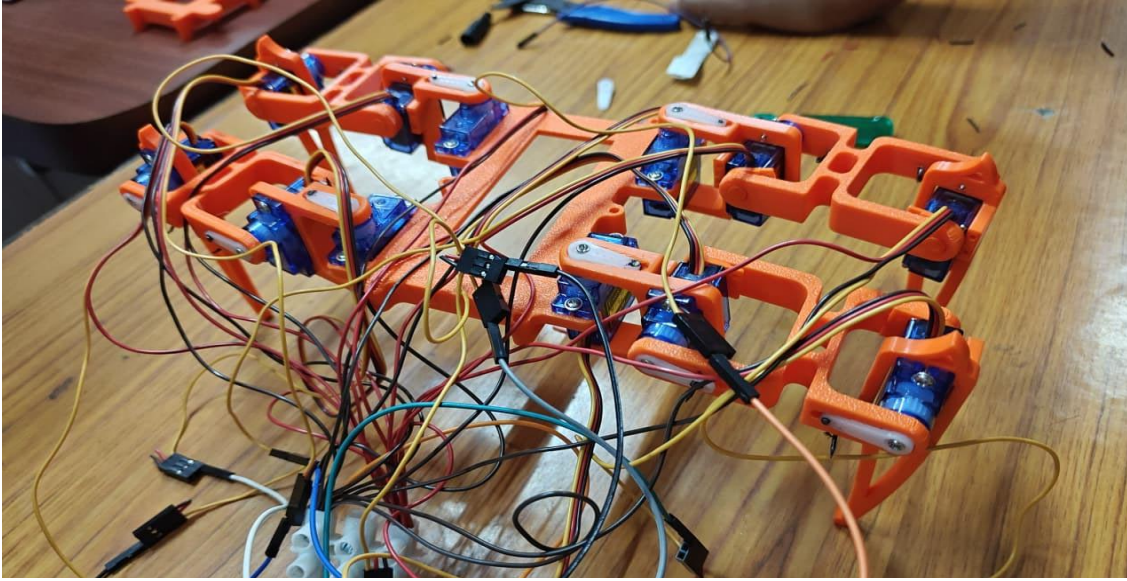
- ❖ The ultrasonic sensor successfully detected nearby obstacles and triggered reactive motion behaviors.
- ❖ The robot halted or redirected its motion before collision, validating the effectiveness of the sensor–controller loop.



Current progress and future scope

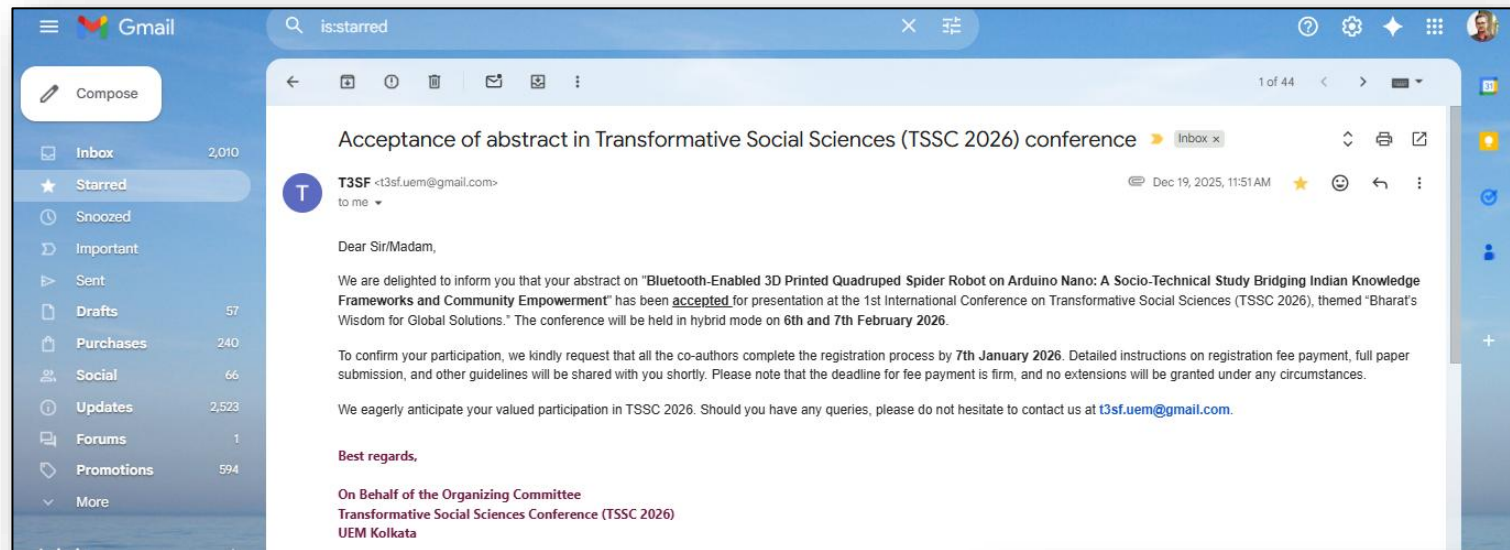
- ❖ **Literature Survey Completed** – Studied existing quadruped robot designs, gait mechanisms, and control techniques.
- ❖ **3D Modeling Finished** – Designed all structural parts of the spider robot using 3D modeling tools for accurate assembly.
- ❖ **Hardware Design Completed** – Built the robot using Arduino Uno, 12 SG90 Servo Motors, 3D printed parts, SMPS (5V), and jumper wires.
- ❖ **Software Development Completed** – Developed control programs using Arduino IDE for smooth servo coordination.
- ❖ **Integrated System Design Achieved** – Successfully combined hardware and software for synchronized and stable robot operation.
- ❖ **Multiple Movement Functions Implemented** – Robot can perform actions like forward walking, backward movement, crawling, dancing, sitting, and handshaking.
- ❖ **Testing and Calibration Completed** – Servos and movements were tested and fine-tuned to improve balance and performance.
- ❖ **Stable and Functional Prototype Ready** – A fully working quadruped spider robot prototype has been successfully developed.
- ❖ **Scope for Future Enhancements Identified** – Design allows future upgrades like sensors, AI-based control, and autonomous navigation.

Pictorial Representation of 3D Model of this project



Publication and Presentation

S. Banerjee, A. Ghosh, S. Bhadra, R. Saha, P. Das, and A. Mukherjee, **“Bluetooth-Enabled 3D Printed Quadraped Spider Robot on Arduino Nano: A Socio-Technical Study Bridging Indian Knowledge Frameworks and Community Empowerment,”** presented at the *1st International Conference on Transformative Social Sciences Conference (TSSC 2026): Bharat's Wisdom for Global Solutions*, Hybrid Mode, Feb. 06, 2026.



- ❖ **Surveillance and Monitoring** – Can be used for simple remote monitoring in restricted areas using wireless control.
- ❖ **Educational Demonstration Tool** – Useful for understanding concepts like servo control, gait movement, and robot coordination in a practical way.
- ❖ **Research Prototype Platform** – Serves as a base model for testing advanced features like obstacle detection, AI-based navigation, and automation.
- ❖ **Entertainment and Demonstration** – Performs actions like walking, crawling, dancing, sitting, and handshaking, making it suitable for exhibitions and demos.
- ❖ **Low-Cost Robotics Development** – Shows how an efficient quadruped robot can be built using affordable components and simple programming.
- ❖ **Future Smart Applications Ready** – Can be upgraded with sensors and modules for real-world applications like inspection, tracking, or autonomous movement.

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Thank You...